

# Mathematics for Computing

## Assignment 1

Poojitha Santhosh

Department of Computer Science and Engineering with Artificial  
Intelligence and Software Engineering

CUSAT

July 31, 2026

- Cardinality of set
- Types of Set in AI and DS
- Examples of Sets in Set Builder form
- Inclusion Exclusion Principle
- Partially Ordered Sets(Posets)
- Hasse Diagram Examples

# Cardinality of a Set

- The **cardinality of a set** is the number of elements present in the set.
- It is represented by the symbol  $|A|$ , where  $A$  is a set.
- For a finite set, cardinality gives the total count of distinct elements in that set.

## Example

Let,

$$A = \{2, 4, 6, 8, 10\}$$

The number of elements in set  $A$  is 5.

Therefore, the cardinality of set  $A$  is:

$$|A| = 5$$

# Types of Sets

## Finite Set

- A **finite set** is a set that contains a fixed and limited number of elements.

$$|A| = n$$

## AI & Data Science Example

- In a machine learning classification model, the output categories form a finite set.
- Example: A sentiment analysis model classifies reviews into:

$$S = \{\text{Positive, Negative, Neutral}\}$$

- Since there are only three possible outcomes,  $S$  is a finite set.

## Set-builder Form

$$A = \{x \in U \mid P(x)\}$$

# Types of Sets

## Infinite Set

- An **infinite set** is a set that contains an unlimited number of elements and cannot be completely counted.

## AI & Data Science Example

- In Data Science, the set of possible real-valued input features in a continuous dataset can be infinite.
- Example: The possible temperature values recorded by an IoT sensor:

$$T = \{x \mid x \in \mathbb{R}, 0 \leq x \leq 100\}$$

- Since temperature values can have infinitely many possible real values,  $T$  is an infinite set.

## Set-builder Form

$$A = \{x \in \mathbb{R} \mid 0 \leq x \leq \infty\}$$

# Types of Sets

## Null Set (Empty Set)

- A **null set** is a set that contains no elements and is represented by  $\emptyset$ .

## AI & Data Science Example

- In a data science dataset, a null set represents a category where no data samples are available.
- Example: In an image classification dataset, if no images belong to the class "Elephant":

$$E = \{\text{Images labeled as Elephant}\} = \emptyset$$

- Since the set contains no elements,  $E$  is a null set.

## General Set-builder Form

$$A = \{x \mid x \in U \text{ and } P(x) \text{ is not satisfied}\}$$

$$A = \emptyset$$

## Countable Set

- A **countable set** is a set whose elements can be listed in a sequence and matched with natural numbers.

## AI & Data Science Example

- In Data Science, the set of possible class labels in a classification model is countable because each category can be listed.
- Example: A handwritten digit recognition model has the classes:

$$D = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

- Since the elements can be counted and listed,  $D$  is a countable set.

## General Set-builder Form

$$A = \{x \mid x = f(n), n \in \mathbb{N}\}$$

# Types of Sets

## Power Set

- A **power set** is the set of all possible subsets of a given set, including the empty set and the set itself.

## AI & Data Science Example

- In feature selection for machine learning, the power set represents all possible combinations of features that can be selected for a model.
- Example: If a dataset has three features:

$$F = \{\text{Age}, \text{Height}, \text{Weight}\}$$

The power set is:

$$P(F) = \{\emptyset, \{\text{Age}\}, \{\text{Height}\}, \{\text{Weight}\}, \{\text{Age}, \text{Height}\}, \{\text{Age}, \text{Weight}\}, \{\text{Height}, \text{Weight}\}, \{\text{Age}, \text{Height}, \text{Weight}\}\}$$

## General Set-builder Form

$$P(A) = \{X \mid X \subseteq A\}$$

# Types of Sets

## Subset

- A **subset** is a set in which every element of one set is also present in another set.

## AI & Data Science Example

- In machine learning, a training dataset is a subset of the complete dataset when only selected samples are used for model training.
- Example:

$$D = \{\text{Image1, Image2, Image3, Image4, Image5}\}$$

$$T = \{\text{Image1, Image2, Image3}\}$$

- Since every element of  $T$  belongs to  $D$ ,

$$T \subseteq D$$

## General Set-builder Form

$$A = \{x \mid x \in B \text{ and } P(x)\}$$

# Types of Sets

## Equal Set

- Two sets are called **equal sets** if they contain exactly the same elements, irrespective of their order.

## AI & Data Science Example

- In machine learning, two feature sets are equal if they contain the same features used for model training.
- Example:

$$A = \{\text{Age, Height, Weight}\}$$

$$B = \{\text{Weight, Age, Height}\}$$

- Since both sets contain the same features,

$$A = B$$

## General Set-builder Form

$$A = \{x \mid P(x)\}$$

# Types of Sets

## Equivalent Set

- Two sets are called **equivalent sets** if they contain the same number of elements, even if their elements are different.

## AI & Data Science Example

- Example:

$$A = \{\text{Image1, Image2, Image3}\}$$

$$B = \{\text{Audio1, Audio2, Audio3}\}$$

- Both sets contain three elements, therefore:

$$n(A) = n(B)$$

$$A \sim B$$

## General Set-builder Form

$$A = \{x \mid P(x)\}$$

# Types of Sets

## Superset

- A **superset** is a set that contains all the elements of another set and is represented as  $A \supseteq B$ .

## AI & Data Science Example

- In machine learning, a complete dataset is a superset of a smaller dataset selected for training.
- Example:

$$D = \{\text{Image1}, \text{Image2}, \text{Image3}, \text{Image4}, \text{Image5}\}$$

$$T = \{\text{Image1}, \text{Image2}, \text{Image3}\}$$

- Since every element of  $T$  is present in  $D$ ,

$$D \supseteq T$$

## General Set-builder Form

$$A = \{x \mid x \in U\}$$

# Types of Sets

## Singleton Set

- A **singleton set** is a set that contains exactly one element.

## AI & Data Science Example

- In machine learning, a singleton set can represent a dataset containing only one unique data sample.
- Example: A facial recognition system identifying one specific registered person:

$$P = \{\text{Person\_ID\_101}\}$$

- Since  $P$  contains only one element, it is a singleton set.

## General Set-builder Form

$$A = \{x \mid x = a\}$$

where  $a$  is the only element of the set.

## 1. Natural Numbers which are Multiples of 3 but not 5

$$A = \{x \in \mathbb{N} \mid x \text{ is a multiple of 3 and } x \text{ is not a multiple of 5}\}$$

Example:

$$A = \{3, 6, 9, 12, 18, 21, \dots\}$$

## 2. Set of Perfect Cubes

$$B = \{x \in \mathbb{N} \mid x = n^3, n \in \mathbb{N}\}$$

Example:

$$B = \{1, 8, 27, 64, 125, \dots\}$$

## 3. Natural Numbers Between 5 and 100

$$A = \{x \in \mathbb{N} \mid 5 < x < 100\}$$

Example:

$$A = \{6, 7, 8, 9, \dots, 99\}$$

## 4. Positive Integers Leaving Remainder 1 When Divided by 5

$$B = \{x \in \mathbb{Z}^+ \mid x = 5n + 1, n \in \mathbb{N}\}$$

Example:

$$B = \{1, 6, 11, 16, 21, \dots\}$$

## 5. Odd Numbers from 1 to 15

$$A = \{x \in \mathbb{N} \mid x = 2n + 1, 1 \leq x \leq 15\}$$

Example:

$$A = \{1, 3, 5, 7, 9, 11, 13, 15\}$$

## 6. Set of Factors of 24

$$B = \{x \in \mathbb{N} \mid x \text{ divides } 24\}$$

Example:

$$B = \{1, 2, 3, 4, 6, 8, 12, 24\}$$

## 7. Ordered Pairs Whose Sum is 5

$$A = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} \mid x + y = 5\}$$

Example:

$$A = \{(0, 5), (1, 4), (2, 3), (3, 2), (4, 1), (5, 0)\}$$

## 8. Numbers Divisible by 2 or 3

$$B = \{x \in \mathbb{N} \mid x \text{ is divisible by 2 or 3}\}$$

Example:

$$B = \{2, 3, 4, 6, 8, 9, 10, 12, \dots\}$$

## 9. Natural Numbers and Their Squares

$$A = \{x^2 \mid x \in \mathbb{N}\}$$

Example:

$$A = \{1, 4, 9, 16, 25, 36, \dots\}$$

## 10. Prime Numbers from 1 to 30

$$B = \{x \in \mathbb{N} \mid x \text{ is prime and } 1 < x \leq 30\}$$

Example:

$$B = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$$

# Inclusion-Exclusion Principle

## General Formula

- The Inclusion-Exclusion Principle is used to find the number of elements in the union of multiple sets by correcting the overcounting of common elements.

For three sets  $A$ ,  $B$ , and  $C$ :

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

## General Formula for $n$ Sets:

$$\left| \bigcup_{i=1}^n A_i \right| = \sum |A_i| - \sum |A_i \cap A_j| + \sum |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n+1} |A_1 \cap A_2 \cap \cdots|$$

# Partially Ordered Sets (Posets)

## Brief Explanation

- A **partially ordered set (Poset)** is a set together with a relation that satisfies three properties:
  - 1 **Reflexive:**  $a \leq a$
  - 2 **Antisymmetric:** If  $a \leq b$  and  $b \leq a$ , then  $a = b$
  - 3 **Transitive:** If  $a \leq b$  and  $b \leq c$ , then  $a \leq c$
- A poset is represented as  $(P, \leq)$ , where  $P$  is a set and  $\leq$  is a partial order relation.

## Example

- Let

$$P = \{1, 2, 3, 6\}$$

with the relation "divides" ( $|$ ).

$$1|2, \quad 1|3, \quad 2|6, \quad 3|6$$

- Therefore,  $(P, |)$  forms a Poset.

# Example of Poset: Hasse Diagram

## Example: Divisibility Poset

Let

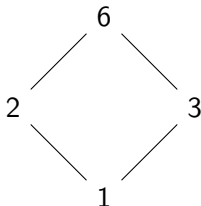
$$P = \{1, 2, 3, 6\}$$

with the relation "divides" ( $|$ ).

The partial order relations are:

$$1|2, \quad 1|3, \quad 2|6, \quad 3|6$$

The Hasse diagram of the Poset  $(P, |)$  is:



# Example of Poset: Hasse Diagram

## Example 2: Subset Poset

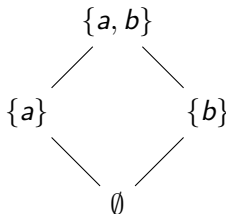
Let,

$$P = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$$

with the relation  $\subseteq$ .

The subset relations are:

$$\emptyset \subseteq \{a\}, \quad \emptyset \subseteq \{b\}, \quad \{a\} \subseteq \{a, b\}, \quad \{b\} \subseteq \{a, b\}$$



$(P, \subseteq)$  is a Poset

# Example of Poset: Hasse Diagram

## Example 3: Factor Poset

Let,

$$P = \{1, 2, 3, 4, 6, 12\}$$

with the relation "divides" ( $|$ ).

The divisibility relations are:

$$1|2, \quad 1|3, \quad 2|4, \quad 2|6, \quad 3|6, \quad 4|12, \quad 6|12$$

